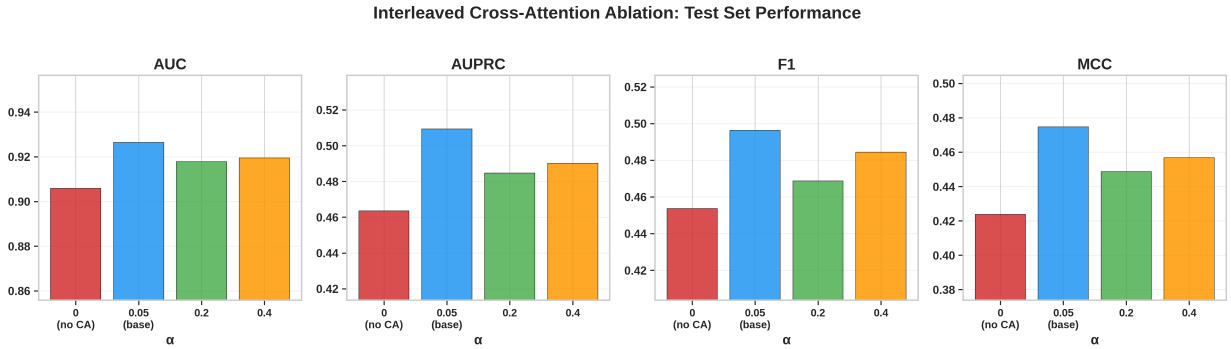


*Figure 1.* EpiFormer performance across 4 MHCA gate initialization values ($\alpha \in \{0.0, 0.05, 0.2, 0.4\}$). The alpha parameter controls the initial cross-attention gating weight; $\alpha$=0 effectively disables encoder cross-attention at initialization. Removing cross-attention ($\alpha$=0) reduces F1 by 0.042 and MCC by 0.051. The base initialization ($\alpha$=0.05) achieves the best performance on all metrics.

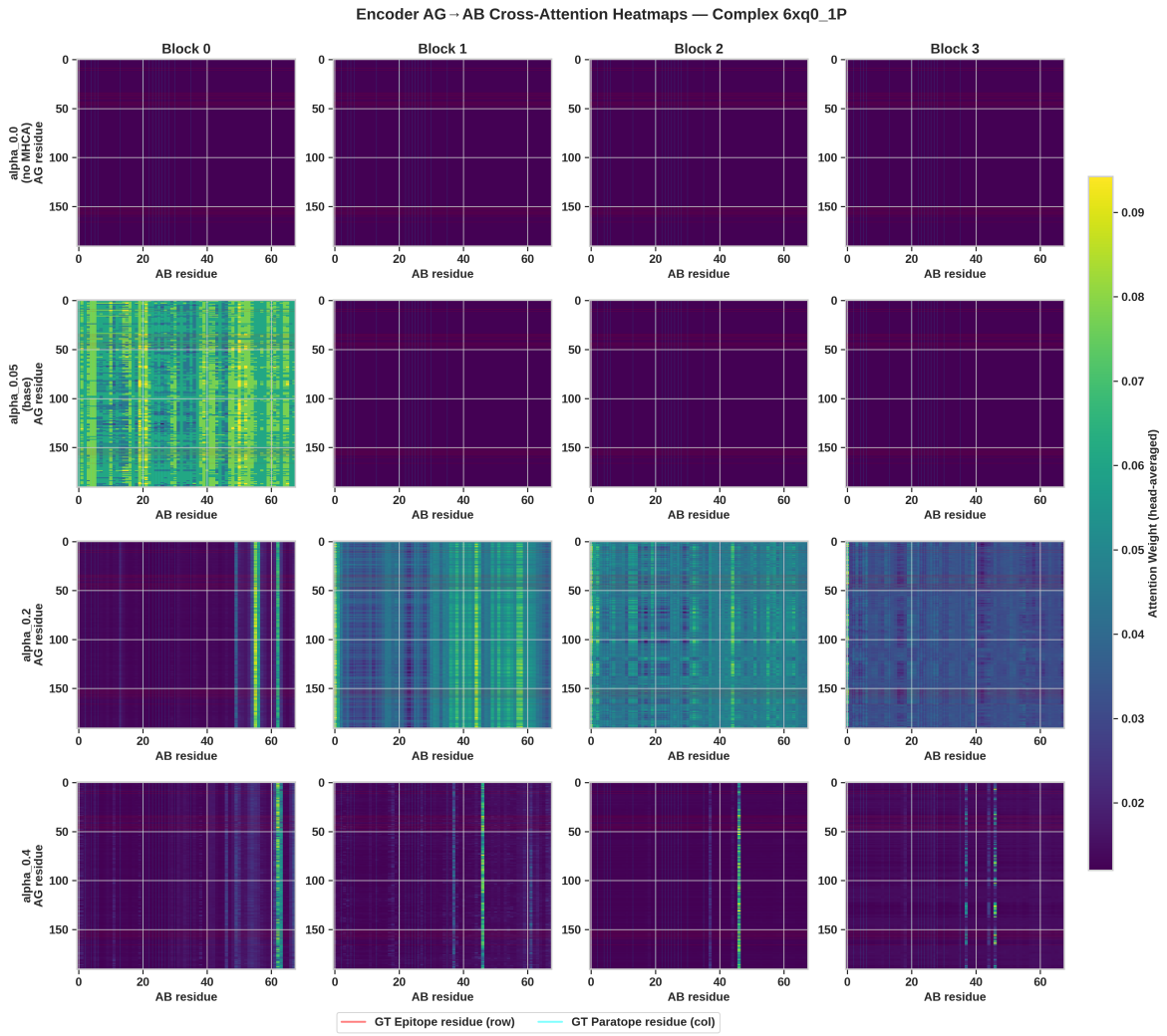


*Figure 2.* Encoder antigen-to-antibody cross-attention heatmaps for 4 MHCA alpha variants ($\alpha \in \{0.0, 0.05, 0.2, 0.4\}$) across 4 encoder blocks, shown for one representative complex from the AsEP epitope_ratio test set. Rows = antigen residues, columns = antibody residues. Ground-truth epitope (red) and paratope (cyan) markers overlaid on axes. Higher alpha initialization values produce stronger vertical banding at CDR positions, indicating more structured cross-chain attention patterns from earlier training stages.

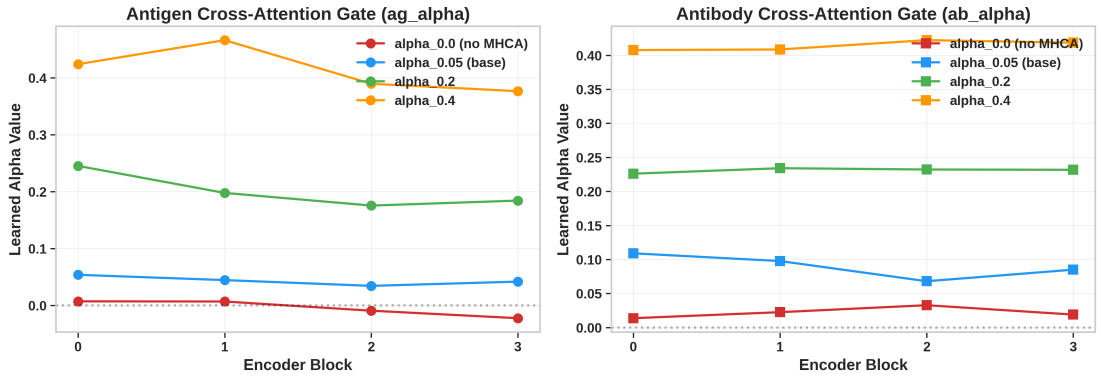


*Figure 3.* Learned alpha gate values across 4 encoder blocks for each MHCA initialization variant ($\alpha \in \{0.0, 0.05, 0.2, 0.4\}$) on the AsEP epitope_ratio split. AB (antibody) alpha consistently exceeds AG (antigen) alpha across all variants and blocks (e.g., 0.109 vs 0.054 in Block 0 for $\alpha$=0.05), reflecting the biological asymmetry that paratope identity depends more heavily on antigen context than vice versa. Even $\alpha$=0.0 models learn non-zero AB gates (0.014–0.033), confirming the model discovers cross-attention utility without explicit supervision.

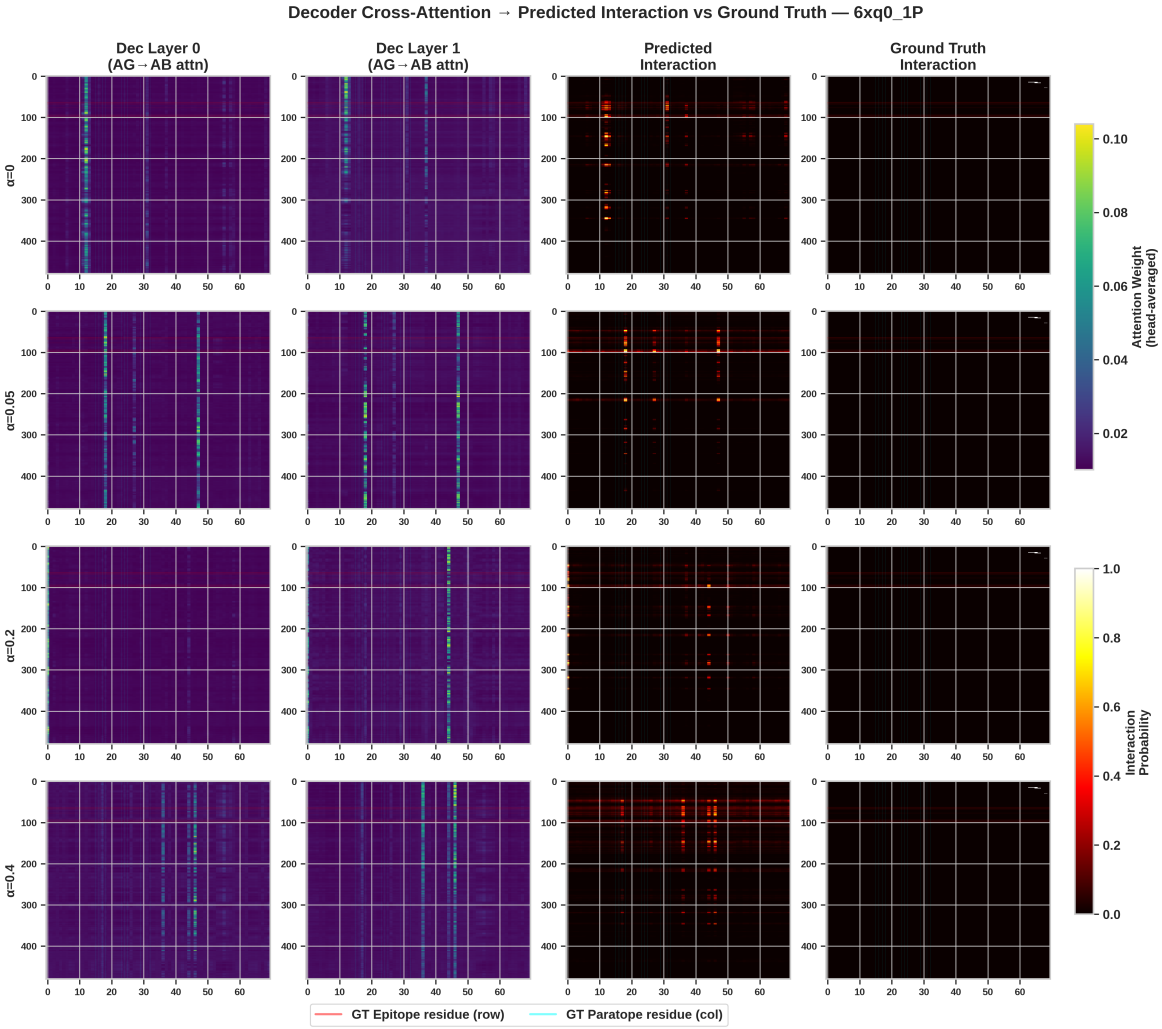


*Figure 4.* Decoder cross-attention weights and predicted interaction maps compared against ground-truth contact maps for 4 MHCA initialization variants ($\alpha \in \{0.0, 0.05, 0.2, 0.4\}$), shown for one representative complex from the AsEP epitope_ratio test set. Decoder attention develops paratope-focused specificity regardless of encoder gate initialization, producing sparse hot spots that align with ground-truth binding edges. Red/cyan lines mark ground-truth epitope/paratope residue positions.